

A 7-vertex tournament whose Sperner capacity exceeds the order of its largest transitive subtournament

JD Jones

October 2026

Abstract

For a digraph R , let $w(R^n)$ be the largest set of words of length n in which every ordered pair of distinct words has a coordinate carrying an arc from the first to the second, and let $C(R) = \lim_n w(R^n)^{1/n}$ be the capacity. For a tournament T , $t(T) \leq C(T)$, where $t(T)$ is the order of its largest transitive subtournament. The equality $C(T) = t(T)$, stated as Conjecture 1.1 in Alon's paper *On the capacity of digraphs* (1998) and attributed there to Körner and Simonyi, was disproved in the same paper; the smallest counterexample given there has 67 vertices, and Alon asked for the smallest size. We show that the answer is seven. The subtournament T_7 of the Paley tournament on 23 vertices induced on $\{0, 1, 2, 3, 9, 14, 18\}$ has $t(T_7) = 4$ and an 18-point transitive clique of its second Sperner power, so $\sqrt{18} \leq C(T_7)$; a rank-5 matrix gives $C(T_7) \leq 5$. Every tournament on at most six vertices has $C = t$, and on seven vertices T_7 is the only exception up to isomorphism. So T_7 is the unique smallest counterexample, and every proper subtournament of it has $C = t$. The statements about T_7 and the Paley tournament on 23 vertices, the general tools, and the classification of tournaments on at most six vertices are proved in Lean 4; the uniqueness on seven vertices and the exact value of the square are on paper, with exact certificates.

1 Definitions and the question

We follow Alon [1]. Let R be a digraph on a finite set V , written $u \rightarrow v$ for an arc. A set $S \subseteq V^n$ is a *clique of the n -th Sperner power* of R if for every ordered pair $u \neq v$ of members of S there is a coordinate i with $u_i \rightarrow v_i$. The pair is ordered: the coordinate for (u, v) and the coordinate for (v, u)

are in general different. Let $w(R^n)$ be the largest size of such a set. The function $n \mapsto w(R^n)$ is supermultiplicative, so the limit

$$C(R) = \lim_{n \rightarrow \infty} w(R^n)^{1/n} = \sup_{n \geq 1} w(R^n)^{1/n}$$

exists and equals the supremum. This is the *capacity* of R , also called its Sperner capacity [6]; some authors use the name for $\log C(R)$ [5, 4]. In Lean the capacity is defined as the supremum, so nothing about limits is assumed.

We also use a one-sided notion. A sequence $u^1, \dots, u^s$ of distinct members of V^n is a *transitive clique* of the n -th Sperner power if for every $i < j$ some coordinate carries an arc from u^i to u^j . Let $\text{tr}(R^n)$ be its largest length. Clearly $w(R^n) \leq \text{tr}(R^n)$. For a tournament T and $u \neq v$, no coordinate carries an arc from u to v exactly when, in every coordinate, $v_i = u_i$ or $v_i \rightarrow u_i$, that is, when $v \rightarrow u$ in the directed strong power. So a transitive clique of the n -th Sperner power of a tournament is the same thing as an acyclic set of its n -th strong power, listed in a topological order.

A tournament T has exactly one arc between any two distinct vertices; it is transitive if its arcs follow a linear order. Let $t(T)$ be the order of the largest transitive subtournament of T . Then $t(T) \leq C(T)$ (Corollary 3 below).

Conjecture 1 (Conjecture 1.1 of [1]). *For every tournament T , $C(T) = t(T)$.*

Alon attributes the conjecture to Körner and Simonyi, citing [3]. The same paper disproves it: Theorem 1.3 there gives, for every n , a tournament on $2n$ vertices with $C \geq \sqrt{n}$ and $t \leq 4 \log_2 n + 2$, and Section 4 there gives an explicit counterexample, the Paley tournament Q_{67} , which has no transitive subtournament on 9 vertices (a computer check reported there) while $C(Q_{67}) \geq \sqrt{67} > 8$. Section 4 of [1] adds that it may be interesting to determine the smallest size of a tournament for which the conjecture fails. On the small cases, [1] argues the conjecture in print for every tournament on at most 4 vertices, from its Theorem 1.2, and reports a verification, with Szabó and Tardos, for at most 5 vertices, with no proof printed. Kiviluoto, Östergård and Vaskelainen [6] determine the Sperner capacity of every digraph on at most 5 vertices except eight, none of which is a tournament. So before this note the least order of a counterexample was known to be at least 6.

Throughout, Q_p is the Paley tournament on $\mathbb{Z}/p$ for a prime $p \equiv 3 \pmod{4}$: $x \rightarrow y$ if and only if $y - x$ is a nonzero square. In Lean the relation on $\mathbb{Z}/23$ is **paley**; the Lean name of each statement is given in brackets.

2 Two tools

Lemma 2 (the lift; `lift_lemma`). *Let $p_0, \dots, p_{N-1} \in V^k$ be distinct, with an arc from p_i to p_j in some coordinate whenever $i < j$. Then for every $m \geq 1$ there is a clique S of the km -th Sperner power of R with $N^m \leq ((N-1)m+1)|S|$.*

Proof. For a word $a = (a_1, \dots, a_m) \in \{0, \dots, N-1\}^m$ let $L(a) \in V^{km}$ be the concatenation $p_{a_1}p_{a_2} \cdots p_{a_m}$. Let $a \neq b$ be two words with $\sum_q a_q = \sum_q b_q$. If $a_q \geq b_q$ for every q , the equal sums force $a = b$; so some q has $a_q < b_q$, and symmetrically some q' has $a_{q'} > b_{q'}$. In block q the point p_{a_q} sends an arc to p_{b_q} in some coordinate, so $L(a)$ sends an arc to $L(b)$ in the corresponding coordinate of V^{km} ; block q' gives an arc from $L(b)$ to $L(a)$. Since the p_i are distinct, L is injective. So every class of words with a fixed index sum is mapped by L to a clique of the km -th power. The index sum takes the $(N-1)m+1$ values $0, \dots, (N-1)m$, so some class has at least $N^m/((N-1)m+1)$ words. $\square$

The proof uses only the forward property for $i < j$. It never composes arcs, and it does not care whether some pairs also carry arcs backward. The lemma is standard in substance: the definition of the capacity through transitive cliques of the powers goes back to Gargano, Körner and Vaccaro [5], and Sali and Simonyi [11] prove the equivalence by a lift with the bound $\text{tr}(R^k)^m/(m+1)^{\text{tr}(R^k)}$ (see also [14, Section 2.2.1]); the equal-index-sum class gives the sharper elementary bound used here, which is the form proved in Lean.

Corollary 3 (`rpow_le_capacity`, `transitiveNumber_le_capacity`). *With $p_0, \dots, p_{N-1}$ as in Lemma 2, $N^{1/k} \leq C(R)$. Hence $\text{tr}(R^k)^{1/k} \leq C(R)$ for every k , and $t(R) \leq C(R)$.*

Proof. By the lemma, $C(R) \geq w(R^{km})^{1/(km)} \geq N^{1/k} ((N-1)m+1)^{-1/(km)}$ for every m , and the last factor tends to 1. The second statement is the first for a longest transitive clique. The third is the case $k=1$, with the vertices of a largest transitive subtournament in their order. $\square$

Since $w \leq \text{tr}$, the corollary shows $C(R) = \sup_k \text{tr}(R^k)^{1/k}$ as well: defining the capacity through transitive cliques, as [6] do, gives the same number. A transitive clique of the k -th power with more than t^k points therefore proves $C > t$.

Lemma 4 (the rank bound; `capacity_le_of_factorization`). *Let $\mathbb{F}$ be a field and $M \in \mathbb{F}^{V \times V}$ a matrix with $M_{vv} \neq 0$ for every v and $M_{uv} = 0$ whenever $u \rightarrow v$. If $r = \text{rank } M$, then $\text{tr}(R^n) \leq r^n$ for every n ; in particular $w(R^n) \leq r^n$ and $C(R) \leq r$.*

Proof. Let $K = M^{\otimes n}$, indexed by V^n : $K_{uv} = \prod_i M_{u_i v_i}$. Then $\text{rank } K = r^n$; equivalently, if $M = AB$ with $A \in \mathbb{F}^{V \times r}$ and $B \in \mathbb{F}^{r \times V}$, then $K = A^{\otimes n} B^{\otimes n}$ factors through $\mathbb{F}^{r^n}$. Let $u^1, \dots, u^s$ be a transitive clique of the n -th power. For $i < j$ some coordinate c has $u_c^i \rightarrow u_c^j$, so $M_{u_c^i u_c^j} = 0$ and $K_{u^i u^j} = 0$; and $K_{u^i u^i} = \prod_c M_{u_c^i u_c^i} \neq 0$. So the $s \times s$ submatrix of K on these points is triangular with a nonzero diagonal, and $s \leq \text{rank } K = r^n$. For a clique the submatrix is diagonal. $\square$

In Lean the hypothesis is stated as a factorization $M = AB$ through $\mathbb{F}^r$, and the conclusion is $C(R) \leq r$. Alon's Theorem 1.2 [1], $C(D) \leq d^- + 1$ for a digraph of maximum in-degree d^- , is the case $M_{uv} = \prod_{j \in N^-(v)} (u - j)$ over $\mathbb{Q}$ with $V = \{1, \dots, k\}$: as functions of u , the columns are polynomials of degree at most d^- , so $\text{rank } M \leq d^- + 1$. Reversing all arcs gives $C(D) \leq d^+ + 1$. The lemma is the rank method of Haemers [7] as adapted to digraphs by Calderbank, Frankl, Graham, Li and Shepp [8] and by Blokhuis [9] (Alon [1] describes their method as a modification of Haemers'); for the Sperner capacity of digraphs it is stated, over the reals, as Theorem 4 of [6], who credit it to [8], and Alon's survey [2] gives an equivalent vector-representation bound over any field, credited to Alon and Tardos. We call the least rank of such a matrix over $\mathbb{F}$ the $\mathbb{F}$ -*minrank* of R . For a tournament, $\text{tr}(T^1) = t(T)$, so the lemma at $n = 1$ shows that the minrank is never below t ; a matrix of rank t proves $C = t$.

Remark 5 (deletion). For any vertex v , $C(T) \leq C(T - v) + 1$. Group a clique of T^n by the set A of coordinates equal to v ; two words of one group agree on A , so their arcs lie outside A , and deleting A leaves a clique of $(T - v)^{n - |A|}$. Hence $w(T^n) \leq \sum_k \binom{n}{k} w((T - v)^{n-k}) \leq (C(T - v) + 1)^n$. We use this only to compare bounds.

3 The tournament T_7

Let $T_7 = Q_{23}[W]$ with $W = \{0, 1, 2, 3, 9, 14, 18\}$ [T7]. The nonzero squares modulo 23 are 1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18, and -1 is not a square, so T_7 is a tournament [T7.isTournament]. Its 21 arcs are

$$\begin{aligned} &0 \rightarrow 1, 0 \rightarrow 2, 0 \rightarrow 3, 0 \rightarrow 9, 0 \rightarrow 18, 1 \rightarrow 2, 1 \rightarrow 3, 1 \rightarrow 9, 1 \rightarrow 14, 2 \rightarrow 3, 2 \rightarrow 14, \\ &2 \rightarrow 18, 3 \rightarrow 9, 9 \rightarrow 2, 9 \rightarrow 18, 14 \rightarrow 0, 14 \rightarrow 3, 14 \rightarrow 9, 14 \rightarrow 18, 18 \rightarrow 1, 18 \rightarrow 3. \end{aligned}$$

The out-degrees of 0, 1, 2, 3, 9, 14, 18 are 5, 4, 3, 1, 2, 4, 2, so the score sequence is (1, 2, 2, 3, 4, 4, 5), and the maximum out- and in-degrees are both 5.

Proposition 6 (`transitiveNumber_T7`). $t(T_7) = 4$.

Proof. 0, 1, 2, 3 is transitive in this order. No five vertices are transitive: a transitive subtournament grows only through common out-neighbours of the vertices chosen so far, and the search over such chains finds none of length 5. The kernel runs the same pruned search. $\square$

Theorem 7 (`T7_square_witness`, `sqrt18_le_capacity_T7`). *The eighteen points*

$$(0, 0), (1, 0), (14, 1), (0, 1), (0, 9), (1, 9), (14, 2), (0, 2), (9, 14), \\ (2, 14), (3, 0), (9, 0), (9, 3), (2, 3), (3, 9), (9, 9), (18, 2), (18, 18)$$

of $W \times W$, in this order, form a transitive clique of the second Sperner power of T_7 . Hence $\sqrt{18} \leq C(T_7)$, and in particular $t(T_7) = 4 < C(T_7)$ [`four_lt_capacity_T7`, `transitiveNumber_lt_capacity_T7`].

Proof. The forward property is a check of the 153 pairs (by `decide` in the kernel). Then Corollary 3 with $k = 2$ and $N = 18$ gives $\sqrt{18} \approx 4.243 \leq C(T_7)$. $\square$

The lift makes the counterexample concrete. With $N = 18$, $k = 2$ and $t^2 = 16$: at $m = 59$ one has $16^{59} (17 \cdot 59 + 1) < 18^{59}$, so the largest index-sum class gives $w(T_7^{118}) > 4^{118}$ [`T7_exists_clique_gt_four_pow`]; and the largest coefficient of $(1 + x + \dots + x^{17})^{38}$ already exceeds 16^{38} , so $w(T_7^{76}) > 4^{76}$. Both are exact integer comparisons (certificate 0001).

Proposition 8. $\text{tr}(T_7^2) = 18$. *There are exactly 84 transitive cliques of size 18 in the second Sperner power of T_7 .*

The lower bound is Theorem 7. The upper bound is proved by the exhaustive search of certificate 0003, a standard-library checker that runs in about ten minutes; it was also obtained by three engines that share no code (a CP-SAT model with positions, a branch and bound in pure Python, and a SAT model with lazy cycle cuts solved by CaDiCaL). The count of 84 is a computation: two of the engines return identical lists. The 17-point set found first in this work is inclusion-maximal but not maximum: no point extends it, and it lies in none of the 84 optimal sets.

Theorem 9 (`capacity_T7_le_five`). $C(T_7) \leq 5$.

Proof. Index rows and columns by 0, 1, 2, 3, 9, 14, 18 and let

$$M = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 0 & 1 \end{pmatrix},$$

so that the rows are $e_0, e_1 + e_{18}, e_1 + e_2 + e_9, -e_2 + e_3 + e_{18}, e_3 + e_9, e_{14}$ and $-e_2 - e_9 + e_{18}$. The diagonal is 1 and every arc position is 0 (compare with the list of arcs). The rows satisfy $v_{18} = v_1 - v_2 = v_3 - v_9$, and the rows at 0, 1, 2, 3, 14 are independent, so $\text{rank}_{\mathbb{Q}} M = 5$; the two relations come from the cyclic triangles $1 \rightarrow 2 \rightarrow 18 \rightarrow 1$ and $3 \rightarrow 9 \rightarrow 18 \rightarrow 3$. Over $\mathbb{F}_2$ the same matrix factors as $M = AB$ with

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} :$$

the rows of B are the rows of M at 0, 1, 2, 3, 14, and the rows at 9 and 18 are $v_1 + v_2 + v_3$ and $v_1 + v_2$. Lemma 4 with $r = 5$ gives the claim. $\square$

Alon's degree bound gives only $C(T_7) \leq 6$. Rank 4 is impossible over every field, since Lemma 4 would give $\text{tr}(T_7^2) \leq 16 < 18$; exhaustive searches over $\mathbb{F}_2$ and $\mathbb{F}_3$ agree. So

$$4.2426\dots = \sqrt{18} \leq C(T_7) \leq 5.$$

Proposition 10. $w(T_7^2) = 7$, and the only clique of size 7 in the second Sperner power is the graph of the anti-automorphism $\varphi = (0\ 3)(1\ 18)(9\ 14)$, which fixes 2.

Proof. If two members $(a, b), (a, b')$ of a clique share a coordinate, both arcs between them must come from the other coordinate, so $b \rightarrow b'$ and $b' \rightarrow b$, which is impossible. So a clique of the second power is the graph of a partial injection and has at most 7 points. A 7-point clique is the graph $\{(a, \pi(a))\}$

of a bijection π , and for $a \rightarrow a'$ the pair needs $\pi(a') \rightarrow \pi(a)$: π is an anti-automorphism. Conversely the graph of an anti-automorphism is a clique. A search over all $7!$ permutations finds $\text{Aut}(T_7)$ trivial and exactly one anti-automorphism, φ ; it is unique because the anti-automorphisms form a coset of $\text{Aut}(T_7)$. $\square$

So T_7 is self-converse with a trivial automorphism group, and the square gives a clique of only 7 points: the separation from t^n appears for transitive cliques at $n = 2$, and for cliques only in a high power. The following are computations, labelled as such.

- $75 \leq \text{tr}(T_7^3) \leq 125$. The lower bound is a witness found by simulated annealing and checked exactly (certificate 0007); the upper bound is Lemma 4. Since $75^2 < 18^3$, the cube does not improve $\sqrt{18}$; a value of at least 77 would.
- $16 \leq w(T_7^3) \leq 19$. The lower bound is a checked witness; the upper bound is a SAT refutation of 20 by two solvers (CaDiCaL and Glucose, with different encodings).
- Fixing one coordinate shows $w(T^n) \leq |V| w(T^{n-1})$, so $w(T_7^4) \leq 133 < 4^4$ and $w(T_7^5) \leq 931 < 4^5$. With $w(T_7) = 1$, $w(T_7^2) = 7$ and $w(T_7^3) \leq 19 < 4^3$, no clique of T_7^n exceeds 4^n for $n \leq 5$, and the least n with $w(T_7^n) > 4^n$ lies in $[6, 76]$.

4 Tournaments on at most six vertices

Theorem 11 (`capacity_eq.transitiveNumber_of_card.le.six`). *Every tournament T on at most six vertices has $C(T) = t(T)$.*

Proof. There are 1, 1, 2, 4, 12 and 56 isomorphism classes of tournaments on one to six vertices. For each class we list a representative T and an $\mathbb{F}_2$ -matrix with diagonal 1, zeros on the arcs and rank $t(T)$ (certificate 0004). Lemma 4 gives $C(T) \leq t(T)$, and Corollary 3 gives $C(T) \geq t(T)$. The list of representatives is complete: the representatives are pairwise non-isomorphic, and $\sum n! / |\text{Aut}(T)| = 2^{n(n-1)/2}$, the number of labelled tournaments on n vertices, for each n . The checker recomputes t , the zero pattern, the ranks over $\mathbb{F}_2$ and both completeness conditions.

The theorem is also proved in Lean, by a different completeness argument that the kernel checks: a tournament on n vertices is encoded by its $\binom{n}{2}$ orientation bits; for each $n \leq 6$ the representatives, with a factorization

AB over $\mathbb{F}_2$ of rank t and a transitive t -chain each, are listed, and the kernel verifies the data and that the bitwise *or* of 2^{code} over every representative and every one of the $n!$ relabellings equals $2^{2^{\binom{n}{2}}} - 1$, so that every orientation is a relabelled representative; the factorization and the chain are then transported along the relabelling. For $n = 6$ this is $56 \cdot 720$ relabellings and about a minute of kernel time. $\square$

For example, the 6-vertex class obtained from Q_7 by deleting a vertex has $t = 3$, degree bound 4, and an $\mathbb{F}_2$ -matrix of rank 3. An exhaustive search over all admissible $\mathbb{F}_2$ -matrices (the $\binom{n}{2}$ entries opposite the arcs are free) shows more: *the $\mathbb{F}_2$ -minrank equals t for every tournament on at most six vertices.* This is a computation, not needed for the theorem.

Corollary 12 (`seven_le_card_of_transitiveNumber_lt_capacity`). *The least order of a tournament T with $C(T) > t(T)$ is 7.*

Proof. Theorem 11 and Theorem 7. $\square$

As a consistency check, two further engines computed $\text{tr}(T^2)$ for every class on at most six vertices; it equals t^2 in every case, as Lemma 4 requires.

5 Seven vertices

Theorem 13. *Of the 456 isomorphism classes of tournaments on seven vertices, 455 have an $\mathbb{F}_2$ -matrix with diagonal 1, zeros on the arcs and rank t , hence $C = t$. The remaining class is that of T_7 .*

This is certificate 0005: the 456 representatives, complete by the same orbit count ($\sum 7!/|\text{Aut}(T)| = 2^{21}$), 455 matrices, and an explicit isomorphism from the remaining representative to T_7 . The values of t over the 456 classes are 3, 4, 5, 6, 7 with multiplicities 1, 120, 287, 47, 1.

Corollary 14. *Up to isomorphism, T_7 is the only tournament on seven vertices with $C > t$, and it is the unique smallest tournament with $C > t$. Since T_7 is self-converse, it is also unique up to reversing all arcs.*

Corollary 15. *Every proper subtournament T' of T_7 has $C(T') = t(T')$. The seven subtournaments on six vertices fall into five isomorphism classes, all with $t = 4$. So T_7 is vertex-minimal with $C > t$.*

Proof. Every proper subtournament has at most six vertices; apply Theorem 11. $\square$

Körner asked ([1], Section 4) for a characterisation of the tournaments T in which every subtournament T' has $C(T') = t(T')$. Corollaries 14 and 15 answer this for tournaments on at most seven vertices: such a tournament has the property if and only if it is not isomorphic to T_7 . Any tournament containing a copy of T_7 lacks it.

Proposition 16. $C(Q_7) = 3$.

Proof. Q_7 has arcs $x \rightarrow y$ for $y - x \in \{1, 2, 4\}$, and $0, 1, 2$ is transitive, so $t(Q_7) \geq 3$. Over $\mathbb{F}_2$ let $M_{uv} = 1$ if $u = v$ or $v \rightarrow u$, and 0 otherwise: the identity plus the converse adjacency matrix. It is the circulant of $f(x) = 1 + x^3 + x^5 + x^6$, and f is divisible by $(x + 1)(x^3 + x + 1)$, a factor of $x^7 - 1$ of degree 4, so $\text{rank}_{\mathbb{F}_2} M \leq 7 - 4 = 3$. Lemma 4 gives $C(Q_7) \leq 3$, and Corollary 3 gives $t(Q_7) = 3 = C(Q_7)$. $\square$

Here the degree bound and the deletion bound both give only 4. Q_7 is the only tournament on seven vertices with $t = 3$. We have not searched the literature for this value beyond the sources of Section 8.

6 The Paley tournament on 23 vertices and its neighbours

T_7 was found inside Q_{23} , which is itself a counterexample with $t = 5$.

Theorem 17. $t(Q_{23}) = 5$, and the 29 points

(22, 7), (8, 2), (8, 3), (8, 11), (1, 13), (14, 13), (1, 6), (5, 6),
 (17, 22), (17, 7), (3, 22), (3, 7), (12, 19), (12, 22), (1, 15), (5, 15),
 (21, 19), (14, 2), (14, 15), (9, 0), (9, 2), (17, 0), (17, 2), (7, 4),
 (3, 8), (7, 6), (9, 8), (21, 8), (11, 8)

form, in this order, a transitive clique of the second Sperner power of Q_{23} . Hence $\sqrt{29} \leq C(Q_{23}) \leq 12$, and $5 < C(Q_{23})$.

Proof. $0, 1, 2, 3, 4$ is transitive. By translation, a transitive 6-set may be assumed to start at 0, and the pruned search over common out-neighbourhoods finds none (the value $t(Q_{23}) = 5$ is in print: [10], as reported in [13], and [12, Table 2]). The forward property is a check of 406 pairs, and Corollary 3 gives the lower bound. Q_{23} is regular of degree 11, so Alon's Theorem 1.2 gives the upper bound 12. $\square$

The Lean names are `paley_isTournament`, `transitiveNumber_paley`, `five_lt_capacity_paley` and `sqrt29_le.capacity_paley`; the upper bound 12 is not formalized.

Alon's anti-diagonal $\{(x, -x)\}$ gives only $w(Q_{23}^2) \geq 23$, that is $C(Q_{23}) \geq \sqrt{23}$, which is below 5. The lift of the 29 points gives $w(Q_{23}^{98}) > 5^{98}$ by pigeonhole at $m = 49$ (certificate 0002).

Two examples smaller than Q_{67} follow from what is in print, although we did not find them stated. The Paley tournament ST_{27} on the field with 27 elements has $t = 5$ by Sanchez-Flores [10], and Alon's anti-diagonal $\{(a, -a)\}$, a clique of the second Sperner power because $x \mapsto -x$ reverses every arc, has 27 points; so $C(ST_{27}) \geq \sqrt{27} > 5$. Removing the point $a = 0$ gives a 26-vertex subtournament with $t \leq 5$ and a 26-point clique, so $C \geq \sqrt{26} > 5$. We re-verified $t = 5$ and both cliques (certificate 0008).

7 What is proved in Lean, and what is on paper

Nineteen theorems are stated in the file `Challenge.lean` of the repository `sperner-t7-lean` and proved in `Solution.lean`, checked by the Lean 4 kernel [15] against Mathlib [16] at tag v4.35.0-rc2, with no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. The definitions are `paley`, `W`, `T7`, `IsTournament`, `IsTransitiveChain`, `transitiveNumber`, `IsSpernerClique`, `spernerCliqueNumber` and `capacity` (the supremum over $n \geq 1$ of $w(R^n)^{1/n}$). The statements are:

- about T_7 : it is a tournament; $t(T_7) = 4$; the 18-point transitive clique of the second Sperner power; some power T_7^n has a clique larger than 4^n ; $4 < C(T_7)$; $\sqrt{18} \leq C(T_7)$; $C(T_7) \leq 5$; $t(T_7) < C(T_7)$;
- general tools, for any digraph on a finite set: the lift (Lemma 2, in natural numbers, with no finiteness assumption); $N^{1/k} \leq C(R)$; $t(R) \leq C(R)$; the rank bound (Lemma 4, for a factorization $M = AB$ through $\mathbb{F}^r$ over any field); $C(R) \leq |V|$;
- about Q_{23} : it is a tournament; $t(Q_{23}) = 5$; $5 < C(Q_{23})$; $\sqrt{29} \leq C(Q_{23})$;
- about small tournaments: every tournament on at most six vertices has $C = t$ (Theorem 11), and a tournament with $t < C$ has at least seven vertices.

Theorem 13, Propositions 8, 10 and 16, the bounds on T_7^3 , and the 26- and 27-vertex examples are on paper. Their finite data are checked by standalone Python programs in `note/certificates/` (standard library only, exact integers, no `assert` statements), each with forged inputs that must fail:

- 0001 the 18-point transitive clique of T_7^2 , $t(T_7) = 4$, and the lift at $n = 118$ and $n = 76$
- 0002 the 29-point transitive clique of Q_{23}^2 , $t(Q_{23}) = 5$, and the lift
- 0003 $\text{tr}(T_7^2) = 18$ exactly, by an exhaustive upper-bound search
- 0004 $C = t$ on at most six vertices: the class lists and the rank- t matrices
- 0005 the 455 rank- t matrices on seven vertices and the isomorphism to T_7
- 0006 the rank-5 matrix for T_7 over $\mathbb{Q}$ and over $\mathbb{F}_2$
- 0007 the witnesses for $\text{tr}(T_7^3) \geq 75$ and $w(T_7^3) \geq 16$
- 0008 $t = 5$ for the Paley tournament on 27 elements and the anti-diagonal cliques

The folder's README gives the re-run line of each certificate. The exact value 18, the count 84, the SAT bound $w(T_7^3) \leq 19$ and the exhaustive minrank searches are computations, and are labelled as such where they are stated.

8 Prior art

As of 2026-10-08 (UTC), we found no tournament on fewer than 67 vertices exhibited in print with capacity above its transitive number, and no statement that every tournament on six vertices has $C = t$. We searched [1] and Alon's survey [2]; the abstract of [3]; [6], whose conclusion leaves six or more vertices to further work, and Vaskelainen's thesis, whose 14-vertex tournament concerns Banks and Slater sets, not capacity; Simonyi's DSc dissertation [14]; the arXiv papers on Sperner capacity (arXiv:math/0312086, 0809.1522, 1309.4930, 1406.0767, 1905.10483, 2312.09224); the Palomar registry; Hexagon; the openai/math repository; and the formal-conjectures repository, zbMATH and Google Scholar. The nearest examples are the 26- and 27-vertex tournaments of Section 6, which follow in one line from [1] and [10]. The two tools of Section 2 are standard in substance ([11], [6]); their values on T_7 and the classification of the small tournaments are what is new.

9 Open questions

1. The exact value of $C(T_7)$, which lies in $[\sqrt{18}, 5]$. A transitive clique of the cube with 77 points would raise the lower bound.
2. The exact values of $\text{tr}(T_7^3) \in [75, 125]$ and $w(T_7^3) \in [16, 19]$.
3. The least n with $w(T_7^n) > 4^n$, which lies in $[6, 76]$.
4. The least order of a counterexample with $t = 5$. By Section 5 it is at least 8, and Q_{23} shows it is at most 23. Computations made during this work, with a single implementation that we have not repeated, found induced subtournaments of Q_{23} on 8 to 11 vertices with a transitive clique of the second Sperner power exceeding t^2 : 17 points for one 8-vertex example with $t = 4$, and 26 points for examples with $t = 5$ on 8, 9, 10 and 11 vertices. If confirmed, the answer is 8. We list these only as leads.
5. The capacity of Q_{23} , which lies in $[\sqrt{29}, 12]$, and of the Paley tournaments in general.

Provenance

Claude (Anthropic) was used throughout; the theorems, proofs, searches and checkers arose in that process, and the repository was written by Claude. The searches used OR-Tools CP-SAT and PySAT with the CaDiCaL and Glucose solvers; every claim that rests on a computation has a standard-library checker or is labelled as a computation. The Lean development was checked by the Lean 4 kernel [15] against Mathlib [16] at tag v4.35.0-rc2. The author directed the project and takes responsibility for the claims. This note is licensed CC BY-SA 4.0; the repository is MIT.

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